

Is Korean’s Higher LLM Token Cost Information or Encoding?

A prospectively frozen code-length measurement on an aligned Korean–English corpus

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Abstract

Aligned Korean text often consumes more billed tokens than English text with parallel content. Two contributions may coexist: the texts can differ in predictive code length, and a billing tokenizer can represent them with different efficiency. Token counts alone cannot distinguish these contributions. We operationalise code length with a fixed language model, $B_M(s) = -\sum_j \log_2 Q_M(z_j \mid z_{<j})$, and compare its corpus sum with content-only token counts. On 7,046 aligned sentence pairs from 278 Global Voices articles, analysed after the held-out plan was prospectively frozen, the Korean side used 1.389 times as many `o200k_base` tokens as the English side (95% cluster-bootstrap CI 1.371–1.406) and had 1.312 times the Qwen3-1.7B-Base-relative code length (1.296–1.328). Their ratio, $R_\eta = R_B/R_T = 0.945$ (0.936–0.953), corresponds to a 5.5% lower model-relative code length per billed Korean token. The primary rule required the one-sided 95% upper bound of R_η to be below 0.95; the observed bound was 0.952, so both the primary rule and the overall all-pass rule failed. The token-ratio criterion passed, and a separate tokenizer-sensitivity control passed: Polyglot-Ko assigned the identical Korean strings 13% fewer token units ($C_K = 0.867$). Exploratory bit-ratio equivalence failed, contradicting a simple encoding-only pattern in which total code lengths are equal. The study therefore demonstrates token-count dependence on tokenizer choice but neither establishes the prespecified 5% efficiency shortfall nor identifies a causal division between linguistic, translation, model, and tokenizer effects.

1 Introduction

Users of LLM application programming interfaces (APIs) are commonly charged per token. Parallel-language studies have shown that the same content can require substantially different token counts, producing unequal monetary and context-window costs [1, 9]. Training-corpus composition is one plausible contributor, but it is not the only one. Korean and English versions are translations rather than identical strings; a translation can expand or contract, and a fixed model can predict the two languages with different accuracy. Token count alone cannot identify these mechanisms.

An information-theoretic comparison puts the two measurements on explicit scales: tokenizer symbols and code bits under a stated predictive model. It does not, by itself, recover intrinsic semantic information or a causal decomposition. If token and model-relative bit ratios move together, the billing-token surplus is accompanied by a similar predictive-code-length surplus. If they separate, tokenizer efficiency is one possible contributor, subject to model and translation confounding.

Related work. Cross-lingual token-count and billing disparities are established empirically [1, 9]. Information-theoretic work has evaluated tokenizers through channel use, entropy, and compression [3, 17]. A recent multilingual token-cost ledger separates removable coding terms from proposed intrinsic terms on FLORES-200 [6]. The narrower contribution here is a Korean–English

case study that combines production-token counts with Qwen-relative sequential code length under a prospectively frozen held-out analysis; it does not claim the first general decomposition or an estimator of intrinsic language entropy.

This paper reports an operational comparison with decision rules fixed in advance. The contributions are: (i) a pooled efficiency ratio R_η that compares Qwen-relative code length with billed-token use; (ii) a pipeline with unit tests that computes it reproducibly with pinned model, tokenizer, and corpus revisions; (iii) a pilot/held-out split with thresholds frozen by content hash before the held-out data were scored; and (iv) the held-out result, which narrowly fails the prespecified primary rule, together with clearly labelled exploratory analyses.

2 Background: tokenizers as codes, models as codebooks

Shannon’s source-coding theorem distinguishes the information in a message from the number of symbols used to write it down [14]. The ideal fractional code length for an event of probability p is $-\log_2 p$ bits; literal prefix-code words have integer lengths, while block and arithmetic coding approach the ideal average. Shannon’s later experiment on printed English estimated the entropy of a language from a human predictor’s guesses [15]; a language model plays the same predictive role today, with the difference that its predictions are explicit probabilities.

Let M be a fixed autoregressive language model with its own tokenizer τ_M . For a message s , write $z = \tau_M(s) = (z_1, \dots, z_n)$ and let c be the fixed document-boundary token. The *model-relative code length* is

$$B_M(s) = - \sum_{j=1}^n \log_2 Q_M(z_j \mid c, z_{<j}), \quad (1)$$

the ideal sequential code length assigned by M ; an arithmetic coder driven by these probabilities approaches this length up to a small finite-coding overhead [5]. No end-of-sequence event is scored, so this is a conditional code length whose decoder is assumed to know the message boundary or content-token count n . Two properties matter. First, $B_M(s)$ does not depend on the *billing* tokenizer: it is computed through τ_M only. Second, it is model-relative surprisal, not intrinsic semantic information. Writing P for the true conditional distribution and Q for the model’s, the expected code length decomposes as $H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q)$, so B_M includes the model’s own prediction error. A model that knows Korean less well than English will overprice Korean text; the direction of this bias is discussed in Section 5.

For a billing tokenizer b with $T_b(s) = |\tau_b(s)|$, its operational efficiency on s is the model-relative code bits per billed token, $\eta_{M,b}(s) = B_M(s)/T_b(s)$. The corpus-level estimands are ratios of sums over aligned pairs i :

$$R_T = \frac{\sum_i T_{i,\text{KR}}}{\sum_i T_{i,\text{EN}}}, \quad R_B = \frac{\sum_i B_{i,\text{KR}}}{\sum_i B_{i,\text{EN}}}, \quad R_\eta = \frac{\sum_i B_{i,\text{KR}} / \sum_i T_{i,\text{KR}}}{\sum_i B_{i,\text{EN}} / \sum_i T_{i,\text{EN}}} = \frac{R_B}{R_T}. \quad (2)$$

As simplified diagnostic patterns, an information-only account predicts $R_B \approx R_T$ and $R_\eta \approx 1$, whereas an encoding-only account predicts $R_T > 1$, $R_B \approx 1$, and $R_\eta < 1$. Real data can contain both contributions, and B_M also contains model mismatch. The quantity $1 - R_\eta$ is the relative shortfall in model-relative code bits per Korean token compared with an English-efficiency counterfactual. It is not a literal count of empty tokens or money that purchased no information. Ratios of sums define a token-weighted corpus estimand and preserve $R_\eta = R_B/R_T$. Means of per-message ratios answer a different, equally-pair-weighted question and can be unstable when denominators are small.

An independent control uses a tokenizer trained on Korean, k . The Korean strings and their Qwen-relative code lengths are unchanged; only the counting tokenizer changes. The absolute

Korean-token reduction

$$C_K = \frac{\sum_i T_{i,\text{KR}}^{(k)}}{\sum_i T_{i,\text{KR}}^{(b)}} \quad (3)$$

measures the token-count change for identical Korean strings under the comparison code. $C_K < 1$ shows tokenizer dependence, but it is not a directly deployable billing counterfactual because a model’s tokenizer cannot normally be replaced without retraining or adaptation. Attributing the difference to training-language balance alone would also require holding the algorithm, vocabulary size, normalisation, and pre-tokenisation fixed, which these two off-the-shelf tokenizers do not.

For completeness, the exploratory relative gap-contraction statistic is $G = R_T^{(k)}/R_T^{(b)}$. It can decrease because the Korean/English ratio narrows by either fewer Korean tokens or more English tokens, so C_K is the direct absolute Korean-token control.

3 Measurement design

3.1 Corpus and split

We use the OPUS packaging of Global Voices v2018q4 in English and Korean [10, 16], a citizen-media news collection whose articles are translated by volunteers. The archive contains 365 article-level alignment groups, of which 347 contain aligned text, yielding 9,017 aligned sentence pairs. Article identity is preserved as a cluster identifier; the corpus has one genre (news), and the translation direction is not recorded.

The release has no train/test division, so a deterministic article-level split was created before any scoring. Articles connected by an identical bilingual pair were merged into indivisible components; components were ranked by a seeded SHA-256 rule (seed 20260910) and the prefix closest to 20% of articles was assigned to the pilot. The pilot contains 69 articles (1,732 pairs) and the held-out set 278 articles (7,285 pairs); no article or exact bilingual pair crosses the split. The files were created, hashed, and mechanically validated during corpus preparation, but no held-out outcome measurements or summaries were examined before the analysis specification was frozen.

Only Unicode NFC normalisation and surrounding-whitespace removal are applied. Validation flags record rows whose Korean side contains no Hangul, rows whose Korean and English text are identical (untranslated photo credits and licence strings), disallowed control characters, and length statistics.

3.2 Estimator model and scoring policy

The estimator is Qwen3-1.7B-Base [11, 12] at revision `ea980cb0`, run in `bfloat16` with the log-softmax computed in `float32`. Every message is scored in evaluation and inference mode from raw next-token logits. The model’s document-boundary token (`<|endoftext|>`, id 151643) is prepended and every content token is scored conditional on it, so that the first content token receives a code length; no end-of-sequence event is charged. Special tokens are never added implicitly (`add_special_tokens=False`), padding is never scored, and the pipeline asserts on every call that the number of scored targets equals the number of content tokens. The observed maxima were 185 Korean and 126 English estimator tokens, far below the 32,768-token context, so no chunking occurred.

3.3 Tokenizers

The baseline billing tokenizer is OpenAI’s `o200k_base` encoding (tiktoken 0.14.0; 199,998 mergeable tokens), used by the GPT-4o generation [8]. The Korean-trained comparison tokenizer is that of EleutherAI’s Polyglot-Ko-1.3B [2, 4] at revision `557e162c`, a 30,000-entry byte-level

BPE trained on a Korean-only corpus. Token counts are content-only, with no chat template or wrapper tokens. Counting tokenizers return integers only; their token ids are never fed to the estimator, and the estimator is scored only through its own tokenizer. The estimator snapshot and baseline tokenizer asset are hash-verified before scoring. The comparison tokenizer is revision-pinned; its resolved files, like those of all three components, are hashed in the run provenance.

3.4 Prospectively frozen analysis

Ratios described as “well above one” or “close to one” are meaningless until thresholds are fixed, and thresholds chosen after seeing the data prove nothing. All thresholds, endpoints, and exclusion rules were therefore written into a specification file whose content hash (SHA-256 850fa9be...) was recorded when it was frozen on 10 September 2026, after the pilot had been analysed and before the held-out set was scored. A pre-results Git commit (2a6d202) provides an additional time-ordered record, but the plan was not deposited in an external pre-registration registry. Confirmatory analysis validates the frozen content hash and refuses pilot-tagged inputs; these internal checks reduce accidental deviation but are not an external cryptographic registration. The frozen decisions are listed in Table 1.

Table 1: Frozen decisions. Bounds are cluster-bootstrap percentiles.

Decision	Value	Rule on the held-out bootstrap
Primary endpoint	R_η	one-sided 95% upper bound $< c_\eta = 0.95$
Token ratio	R_T	one-sided 95% lower bound $> c_T = 1.20$
Tokenizer control	C_K	one-sided 95% upper bound $< c_C = 0.90$
Overall rule	all three confirmatory comparisons must pass	global conjunction: all three pass
Bit-ratio equivalence (exploratory)	R_B	90% CI inside [0.8333, 1.20]
Exclusions	identical KR/EN text; no Hangul on KR side	applied identically to pilot and held-out
Bootstrap	10,000 replicates, seed 20260910	whole articles resampled, both languages together

The margins operationalise effects judged practically material for this study: at least 20% extra billed tokens ($c_T = 1.20$), at least a 5% efficiency shortfall ($c_\eta = 0.95$), and at least a 10% Korean-token reduction under the comparison tokenizer ($c_C = 0.90$). They are researcher-chosen margins, not externally standardised cutoffs. The exploratory bit-ratio equivalence interval allows a reciprocal 20% departure from equality. The pilot informed feasibility and precision. Its point estimates ($R_\eta = 0.940$, $R_T = 1.377$, $C_K = 0.870$ after exclusions) located the effect, and a precision simulation, which resampled the pilot’s per-article sums to the held-out article count, described the distribution of the bootstrap bounds that the rules would use. For $c_\eta = 0.95$ that simulation gave a pass probability of 0.78 conditional on the pilot values carrying over unchanged; the threshold lay 1.25 pilot standard errors above the pilot point estimate. This is recorded here because it bears directly on the interpretation of the confirmatory outcome.

The frozen confirmatory claim was the conjunction $R_\eta < 0.95$, $R_T > 1.20$, and $C_K < 0.90$ under their stated one-sided bounds. Because R_B equivalence was exploratory, this design does not confirmatorily adjudicate the full simple encoding-only prediction $R_B \approx 1$.

3.5 Statistical analysis

All estimands are ratios of corpus sums, Eq. (2). Uncertainty is quantified by a paired, stratified cluster bootstrap: articles are treated as exchangeable draws and resampled with replacement within genre strata, every pair of a resampled article enters with both its Korean and English measurements, and each pooled ratio is recomputed in every replicate. Because the estimands are functions of column sums only, per-article sums are precomputed and the result is computationally identical to row-level resampling of whole articles. The resulting intervals target a superpopulation of Global Voices-style Korean–English volunteer-translated citizen-media/news articles, not all Korean and English use. The identity $R_\eta = R_B/R_T$ is asserted in every replicate. One-sided decision bounds are the 5th or 95th percentiles; equivalence for R_B uses the two one-sided tests procedure, requiring the 90% percentile interval to lie inside the pre-specified interval [13]. The all-pass rule is one global conjunction; every component must pass, so no multiplicity correction is invoked for that global claim.

Exploratory analyses, which never feed the confirmatory rule, are: pooled estimands within terciles of English character count; the distribution of per-message ratios; and paired cluster-level sign-flip permutation tests in which the Korean and English measurements of all pairs in a randomly chosen set of articles are exchanged, with the log pooled ratio as statistic and 10,000 permutations. These tests address the sharp null that language labels (or, for C_K , tokenizer labels) are exchangeable within article, which is stronger than the statement that a pooled ratio equals one. The three language ratios are functionally dependent and are reported as one finding seen from three angles.

3.6 Implementation

The pipeline is a Python package (`klen`) with 66 unit tests, including hand-verifiable code-length cases (a uniform fake model yields exactly $\log_2 10$ bits per token; a biased fake model exactly 1 bit), whole-cluster resampling verified against the enumerable set of possible replicates, stratum preservation, freeze and tamper detection, and an independent re-implementation cross-check against a tiny random model. Provenance for every run records package versions (PyTorch 2.14.0, Transformers 5.17.0, Python 3.13.1), tokenizer file hashes, the dataset hash, the platform (`arm64`) and device backend (`mps`), the specification, and the validation report.

4 Results

4.1 Sample

After the frozen exclusion rules, the held-out set comprises 7,046 pairs in 278 articles (239 rows dropped: 104 with identical Korean and English text and a further 135 with no Hangul on the Korean side). The pilot after the same rules comprises 1,672 pairs in 69 articles. No row changed under NFC normalisation; no pair id was duplicated.

4.2 Pooled estimands and confirmatory decisions

Table 2 gives the held-out estimates. The Korean side costs 38.9% more `o200k_base` tokens and has 31.2% greater Qwen-relative code length than the English side. The efficiency ratio is $R_\eta = 0.945$, corresponding to 5.5% lower Qwen-relative code bits per billed Korean token, with a 95% interval of 4.7% to 6.4%. The token-ratio and tokenizer-sensitivity component rules pass their prespecified bounds. The primary comparison does not: the one-sided 95% upper bound of R_η is 0.9519, above the frozen threshold of 0.95, and under the overall rule the confirmatory claim is therefore not supported. The bit ratio is outside the exploratory equivalence interval; the simple encoding-only pattern $R_B \approx 1$ is not observed.

Table 2: Held-out pooled estimands (278 articles, 7,046 pairs) with 95% cluster-bootstrap percentile intervals and prespecified decisions. R_T counts o200k_base tokens. C_K is the Polyglot-Ko/o200k_base ratio for the same Korean strings. Bits are Qwen3-1.7B-Base-relative code lengths.

Quantity	Point	95% CI	Decision bound	Result
R_T token ratio KR/EN	1.3888	[1.3714, 1.4058]	lower 1.3741 > 1.20	pass
R_B bit ratio KR/EN	1.3121	[1.2959, 1.3278]	90% CI [1.2984, 1.3252], outside [0.8333, 1.20]	no equivalence*
$R_\eta = R_B/R_T$ (primary)	0.9448	[0.9363, 0.9532]	upper 0.9519 $\not\leq$ 0.95	fail
$1 - R_\eta$ shortfall	0.0552	[0.0468, 0.0637]	—	—
C_K Korean tokens, Polyglot/o200k	0.8665	[0.8601, 0.8729]	upper 0.8719 < 0.90	pass
R_T under Polyglot-Ko	0.5095	[0.5025, 0.5163]	—	exploratory
G gap contraction	0.3668	[0.3629, 0.3707]	—	exploratory
Overall (all three criteria)				fail

*Exploratory only; not part of the overall confirmatory rule.

The pilot and held-out estimates agree closely (Table 3). The held-out R_η is 0.0048 above the pilot value, about 1.1 held-out bootstrap standard deviations; with the threshold placed 1.25 pilot standard errors above the pilot estimate, that shift was enough.

Table 3: Pilot versus held-out point estimates after identical exclusions.

Quantity	Pilot (69 articles, 1,672 pairs)	Held-out (278 articles, 7,046 pairs)
R_T	1.3769	1.3888
R_B	1.2943	1.3121
R_η	0.9400	0.9448
C_K	0.8697	0.8665

4.3 Exploratory analyses

Message length. Table 4 splits the held-out pairs into terciles of English character count. The token surplus, the bit surplus, and the efficiency shortfall all shrink with length: $1 - R_\eta$ is 9.3% in the shortest tercile and 4.4% in the longest; their marginal intervals do not overlap, but no direct between-tercile contrast was tested. Because pooled ratios weight by tokens, the corpus-level estimate is dominated by long messages. Each message includes one boundary-conditioned first-token term, which occupies a larger share of a short message, but no alternative-BOS sensitivity analysis was run; the cause of the pattern is therefore unresolved.

Table 4: Held-out pooled estimands within terciles of English character count (exploratory). Articles may span terciles; bootstrap within tercile.

Tercile	Chars (EN)	Pairs	R_T	R_B	R_η [95% CI]	C_K
Q1 (short)	[3, 74]	2,349	1.564	1.419	0.907 [0.894, 0.921]	0.882
Q2	(74, 137]	2,348	1.415	1.318	0.931 [0.919, 0.944]	0.868
Q3 (long)	(137, 587]	2,349	1.329	1.271	0.956 [0.946, 0.966]	0.861

Per-message distribution. Per-message ratios are descriptive only. The median per-pair token ratio is 1.429 (5th–95th percentiles 0.762–2.364), the median bit ratio 1.357 (0.766–2.091), and the median efficiency ratio 0.944 (0.666–1.369). Their arithmetic means (1.489, 1.396, 0.973)

happen to exceed the pooled estimates here. These means are equal-pair-weighted descriptive summaries, not biased estimates of the token-weighted pooled quantities.

Permutation tests. Exchanging the Korean and English measurements of whole articles 10,000 times gives observed log pooled ratios of $+0.328$ (R_T), $+0.272$ (R_B), -0.057 (R_η) and -0.143 (C_K), each with two-sided $p = 1/10,001$, the smallest attainable value. Under the sharp label-exchangeability null, every observed statistic is in the extreme permutation tail. These exploratory tests are not evidence for the prespecified effect sizes and are not the basis of the confirmatory decisions.

4.4 Post hoc: the baseline tokenizer generation

To understand why the effect is small, we recomputed R_T and R_η on the held-out set holding the Qwen3 bits fixed and substituting the two previous OpenAI encodings for the billing tokenizer (Table 5). This was done after the confirmatory analysis and is not part of it.

Table 5: Held-out R_T and R_η under three OpenAI encodings with the same Qwen3-1.7B-Base bits (post hoc, exploratory).

Encoding	Associated model family	R_T	R_η
r50k_base	GPT-3	4.591	0.286
cl100k_base	GPT-3.5 / GPT-4	2.238	0.586
o200k_base	GPT-4o (frozen)	1.389	0.945

The model-relative efficiency shortfall is 71% with r50k_base, 41% with cl100k_base, and 5.5% with o200k_base. For these same strings, the sequence is consistent with more efficient Korean token allocation in newer OpenAI encodings. This post hoc comparison neither identifies vocabulary size or multilingual training as the cause nor reconstructs historical model or API costs.

5 Discussion

5.1 Why the primary comparison failed

Two distinct things happened. Statistically, the threshold was set 1.25 pilot standard errors above the pilot estimate, the pre-freeze simulation gave the rule only a 78% chance of passing even if the pilot value were exactly right, and the held-out estimate moved up by an unremarkable 1.1 bootstrap standard deviations. None of this justifies revising the threshold after observing the result; that constraint is the purpose of freezing the rule.

Substantively, R_η is close to one because R_B is large, not because R_T is small. Under the current tokenizer the Korean token surplus is 39%, and the model spends 31% more bits on the Korean side. Three contributors, which the present design cannot separate, push R_B upward:

1. *Estimator representation.* Bits are summed over the estimator’s own tokens, and Qwen3’s tokenizer assigns the Korean side 1.554 times as many tokens as the English side. Tokenization defines the model’s factorisation, but finer segmentation alone does not mathematically imply a larger total chain-rule code length. This diagnostic therefore does not identify how much of R_B comes from the representation.
2. *Model fluency.* The D_{KL} term is larger for the language the model predicts less well. Qwen3 reports multilingual training, but the public model report does not disclose the language proportions needed to calibrate a Korean–English KL-gap difference. The held-out code lengths are 4.87 bits per Qwen token for Korean versus 5.77 for English, and 1.42 versus

1.23 bits per UTF-8 byte; both are unit-dependent diagnostics rather than a calibration of fluency.

3. *Translation content.* Global Voices Korean articles are volunteer translations that may add explanation; parallel meaning does not guarantee equal information.

The model-mismatch and translation terms can alter R_B , but their magnitudes and net direction are not measured here. Consequently, the 5.5% shortfall is not a lower bound on a causal encoding effect.

5.2 What the evidence supports

The prespecified claim of at least a 5% Korean shortfall in model-relative code bits per billed token was not established at the 95% level. Within the stated corpus and estimators, the evidence supports a 38.9% Korean `o200k_base` token surplus and passage of its 20% component criterion. It also supports tokenizer sensitivity: Polyglot-Ko assigned 13.3% fewer Korean token units and passed the 10% component criterion. However, the exploratory R_B equivalence prediction failed, the primary R_η rule failed, and therefore the compound confirmatory hypothesis failed. The historical-tokenizer pattern is descriptive rather than causal.

5.3 Limitations

The estimator is a single small model; B_M is Qwen3-1.7B-relative surprisal, not semantic information, and its cross-language KL gap is unquantified. Omitting EOS makes it a code length conditional on external message framing. Content-only token counts also omit chat wrappers and other API-specific overhead. A second estimator and the multi-genre FLORES+ corpus [7], for which a specification file is prepared, remain exploratory replications.

The corpus contains one genre with unknown translation direction and predates the model’s release, so training-data contamination cannot be ruled out. The bootstrap generalises only to Global Voices-style translated articles; repeated content linking a small number of articles may also make article-only intervals slightly optimistic. The three language ratios are functionally dependent. C_K shows token-count sensitivity but neither its cause nor a tokenizer that can be swapped into an existing billed model. Historical encodings and R_B equivalence are exploratory. Most importantly, the present held-out outcomes have now been opened: any rerun or revised threshold on this corpus is exploratory. A new confirmatory analysis would require genuinely new, untouched data frozen under a new plan.

6 Conclusion

The overall confirmatory hypothesis failed because its primary efficiency criterion failed. On 7,046 held-out aligned pairs, Korean used 38.9% more `o200k_base` tokens and had 31.2% greater Qwen3-relative code length, giving an estimated 5.5% efficiency shortfall (95% CI 4.7–6.4%). The prespecified rule required the one-sided upper bound to show a shortfall strictly greater than 5%, and that bound corresponded to 4.81%. A separate tokenizer-sensitivity criterion passed because Polyglot-Ko assigned the same Korean strings 13.3% fewer token units. Tokenizer choice therefore demonstrably changes token counts, but this study did not establish the prespecified efficiency magnitude or identify what causal share of the Korean token surplus belongs to translation, model fit, or tokenizer design.

Reproducibility

Code, tests, the frozen specification, and confirmatory analysis outputs (bootstrap replicates, provenance, and per-pair measurements with corpus text removed) are in the [archived project revision](#). The corpus is rebuilt from a hash-verified OPUS archive by a script that reproduces the split. The estimator snapshot and baseline tokenizer asset are verified before scoring; all resolved tokenizer files and revisions are recorded in the run provenance. The historical-tokenizer comparison was computed post hoc and is not part of the archived confirmatory outputs.

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A Pilot precision simulation

Before the thresholds were chosen, the pilot’s 69 per-article sums were resampled with replacement to 278 articles 400 times; within each simulated sample a 2,000-replicate cluster bootstrap produced the bounds that the confirmatory rules use. Table 6 reports the quantiles of those bounds across simulations. The observed upper bound, 0.9519, lies near the upper tail (about the 90th empirical percentile) of this simulated distribution and below its 95th percentile, 0.9530.

Table 6: Quantiles across 400 simulated held-out samples (278 articles) of the decision bounds, using pilot per-article sums as the population.

Bound	5%	25%	50%	75%	95%
R_T lower 95%	1.3325	1.3443	1.3518	1.3586	1.3684
R_B 90% CI low	1.2563	1.2654	1.2723	1.2782	1.2865
R_B 90% CI high	1.2863	1.2947	1.3013	1.3067	1.3157
R_η upper 95%	0.9399	0.9440	0.9467	0.9495	0.9530
C_K upper 95%	0.8853	0.8902	0.8933	0.8970	0.9030

B Exclusion sensitivity on the pilot

Before the exclusion rules were chosen, every candidate rule was applied to the pilot. Dropping identical-text rows, rows without Hangul, rows shorter than five characters, rows with a Korean/English character ratio outside [0.2, 1.5], or all of these together moved R_η within 0.938–0.941 and R_T within 1.368–1.393. The pooled estimands are insensitive to these choices; the rules adopted remove untranslated material on grounds of validity rather than effect.